

Business Analytics and Data Visualization

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UNIT 1 — Analytics Process and Data Quality in BFSI

Total: 9 hrs (Classroom Teaching: 7 hrs, Self Learning: 2 hrs)

Syllabus coverage of this document:

- Business analytics — definition, types and applications in BFSI
- Data analytics lifecycle — problem definition, data collection, processing, analysis, visualisation and decision
- Types of analytics — descriptive, diagnostic, predictive and prescriptive
- Data sources in BFSI — transactional data, market data, customer data and regulatory data
- Data quality dimensions — accuracy, completeness, consistency, timeliness and validity
- Common data quality issues in BFSI — missing values, duplicates, outliers and format inconsistencies
- Data governance and data management frameworks in financial services

BFSI = Banking, Financial Services and Insurance. Throughout this unit, every concept is explained point-wise and then tied to a real Indian BFSI situation so that the theory becomes usable in a dashboard or report.

1. Business Analytics — Definition

- **Definition:** Business analytics (BA) is the systematic use of data, statistical methods, technology and domain knowledge to convert raw business data into insight that drives a decision or an action.
- **Key idea:** Data alone is not analytics. Analytics = Data + Method + Business question + Decision.
- **Three ingredients:** (a) **Data** — transactions, customers, markets; (b) **Tools & techniques** — Excel, SQL, Tableau, Power BI, statistics, machine learning; (c) **Domain context** — how a bank, insurer, AMC or FinTech actually earns money and takes risk.
- **Difference from reporting:** Reporting says “what happened”; analytics also answers “why it happened, what will happen and what should we do”.
- **Difference from Business Intelligence (BI):** BI is largely the infrastructure and dashboards that present data; analytics is the reasoning layer that produces recommendations. In practice BI is a subset of BA.
- **Data → Information → Insight → Decision chain:** 10 lakh card swipes (data) → “spend rose 12% MoM” (information) → “growth is driven by young salaried users on e-commerce” (insight) → “launch a cashback offer for that segment” (decision).

Real-life example: An Indian bank has 4 crore savings accounts. Raw balances mean nothing by themselves. Analytics groups accounts by balance band, salary credit pattern and city; finds that 8%

of accounts hold 61% of deposits; and recommends a dedicated relationship manager programme for that 8%. That recommendation — not the table — is business analytics.

2. Types of Business Analytics (Overview)

Analytics maturity is normally described on a four-step ladder. Value and difficulty both increase as you move down.

Type	Question answered	Typical methods	BFSI example
Descriptive	What happened?	Aggregation, KPIs, dashboards, trend charts	Total NPA of the bank in Q1 was ₹2,400 crore, up from ₹2,100 crore
Diagnostic	Why did it happen?	Drill-down, slice-and-dice, correlation, root-cause, cohort analysis	NPA rise is 80% from unsecured personal loans issued in 2 zones during a festive push
Predictive	What will happen?	Regression, classification, time series, ML	Model gives each borrower a probability of default in the next 6 months
Prescriptive	What should we do?	Optimisation, simulation, decision rules, what-if scenarios	Cut sanction limits by 20% in those zones, raise pricing by 150 bps for high-risk scores

- **Cognitive / AI-assisted analytics** is sometimes added as a fifth step — natural language generation writes the commentary of the dashboard automatically (covered in Unit 5).
- **Sequence matters:** never jump to predictive models before descriptive and diagnostic work is clean; bad data quality makes advanced models confidently wrong.

3. Applications of Business Analytics in BFSI

3.1 Banking

- **Credit scoring and underwriting:** approve/decline and price a loan from bureau score, income, banking behaviour. Example — CIBIL score plus salary-credit stability drives a personal loan decision.
- **Fraud detection:** real-time rules and models on card and UPI transactions. Example — a card used in Indore at 10:00 and in Dubai at 10:20 is flagged instantly.
- **Collections analytics:** rank delinquent accounts by probability of recovery so agents call the most recoverable first.
- **Cross-sell and next-best-offer:** a customer with a 3-year salary account and no credit card gets a pre-approved card offer.
- **Branch and ATM optimisation:** footfall and cash withdrawal patterns decide cash loading and branch timings.
- **Churn / attrition analysis:** falling average balance plus stopped salary credit predicts account dormancy.
- **Regulatory and risk reporting:** CAR, liquidity, stress testing reports for RBI.

3.2 Insurance

- **Risk-based pricing (actuarial analytics):** motor premium varies by vehicle age, city, claim history.
- **Claims fraud analytics:** clusters of claims from the same garage/hospital or same agent are flagged.
- **Persistency and renewal analytics:** predict which life-insurance policies will lapse in the 13th month and send reminder campaigns.
- **Reserving and solvency monitoring:** is the insurer holding enough capital for future claims (IRDAI solvency margin)?
- **Customer lifetime value:** decide service level for high-value policyholders.

3.3 Asset Management / Wealth

- **Portfolio performance analytics:** returns, alpha, beta, Sharpe ratio, tracking error of a fund.
- **AUM growth and flow analysis:** which schemes and which distributors bring net inflows.
- **Risk analytics:** VaR, drawdown, concentration limits.
- **Robo-advisory:** map a risk-profile questionnaire to a model portfolio automatically.

3.4 FinTech

- **Growth analytics:** DAU/MAU, funnel conversion from app install to first transaction.
- **Unit economics:** CAC vs LTV per acquisition channel.
- **Alternative-data credit scoring:** UPI cash-flow, GST returns and app data for thin-file borrowers (BNPL, small-business lending).
- **Personalisation:** in-app offers based on real-time behaviour.

Real-life example: A payments FinTech notices in its dashboard that MAU is growing 10% but revenue is flat. Diagnostic drill-down shows new users are only doing zero-margin utility bill payments, not merchant payments. Prescriptive action: change onboarding to push merchant QR payments. This one insight cycle uses all four types of analytics.

4. The Data Analytics Lifecycle

The lifecycle is the standard end-to-end process of an analytics project. It is iterative — findings at any step can send you back to an earlier step.

Step	What is done	Deliverable	BFSI illustration (car-loan default project)
1. Problem definition	Convert business pain into a measurable analytical question; define scope, success metric, stakeholders	Problem statement + KPI definition	"Reduce 90+ DPD in car loans from 3.2% to 2.5% within 2 quarters"
2. Data collection	Identify and pull required internal + external data; check permissions and privacy	Raw dataset / data inventory	Loan master, EMI repayment history, bureau score, dealer code, customer KYC

Step	What is done	Deliverable	BFSI illustration (car-loan default project)
3. Data processing (preparation)	Clean, de-duplicate, treat missing values and outliers, standardise formats, join tables, create derived features	Analysis-ready table	Standardise dates to DD-MM-YYYY, remove duplicate loan IDs, create "EMI/income ratio"
4. Analysis / modelling	Descriptive stats, hypothesis testing, segmentation, predictive model, validation	Model findings +	Logistic regression giving default probability; AUC 0.78; top driver = EMI/income > 55%
5. Visualisation and communication	Build dashboard / report, choose right charts, write the insight story	Dashboard + one-page insight summary	Power BI page: portfolio at risk by dealer, vintage curve, risk-band heatmap
6. Decision and action	Business takes the action; changes are implemented in policy or systems	Policy change / campaign	Cap loan-to-value at 80% for dealers in the worst risk quartile
7. Monitoring and feedback	Track whether the KPI actually moved; refresh data and re-train	Monitoring dashboard, model drift report	90+ DPD tracked monthly; model re-validated every quarter

- **Effort reality:** roughly 60–70% of project time goes into steps 2 and 3 (collection and cleaning), not modelling.
- **Common failure:** skipping step 1. A dashboard built without a defined business question becomes "chart junk" nobody uses.
- **Related standard frameworks:** CRISP-DM (Business understanding → Data understanding → Data preparation → Modelling → Evaluation → Deployment) and SEMMA (Sample, Explore, Modify, Model, Assess) describe the same cycle with different names.

5. Data Sources in BFSI

5.1 Transactional data

- **Meaning:** record of every financial event — debit/credit entries, card swipes, UPI/NEFT/IMPS transfers, EMI payments, premium receipts, claim payouts, trades.
- **Characteristics:** very high volume, timestamped, generated by core systems (core banking, card switch, policy admin system), append-only and audit-critical.
- **Analytical uses:** spend analysis, fraud detection, cash-flow based credit scoring, liquidity forecasting, CASA behaviour.
- **Example:** UPI in India processes over a billion transactions a day — each row has payer VPA, payee VPA, amount, timestamp, status.

5.2 Market data

- **Meaning:** externally generated prices and rates — equity prices, NAVs, bond yields, repo rate, FX rates, commodity prices, index levels, volatility.
- **Sources:** NSE/BSE, RBI, Bloomberg, Refinitiv, AMFI, CCIL.
- **Characteristics:** time-series, high frequency (tick-level to daily), needs corporate-action adjustment, licensed/paid.
- **Uses:** portfolio valuation, mark-to-market, VaR, benchmark comparison, treasury and ALM decisions.
- **Example:** a mutual fund computes daily NAV using closing prices; a tracking-error report compares fund return with Nifty 50 return.

5.3 Customer data

- **Meaning:** who the customer is and how they behave — KYC demographics (name, age, PAN, address), income, occupation, relationship holdings, channel usage, complaints, call-centre logs, app clickstream, consented bureau and Account Aggregator data.
- **Characteristics:** contains PII and is regulated (DPDP Act 2023), often duplicated across systems, slowly changing.
- **Uses:** segmentation, cross-sell, churn prediction, service personalisation, single customer view (Customer 360).
- **Example:** the same person exists as a savings customer, a credit-card holder and an insurance policyholder in three systems — building one Customer-ID view is a core BFSI analytics project.

5.4 Regulatory data

- **Meaning:** data prepared and submitted to regulators, and data required to prove compliance — RBI ADF/CIMS returns, Basel III capital and CAR reporting, IRDAI returns, SEBI filings, AML/CFT suspicious-transaction reports, FATCA/CRS, CKYC records, credit-bureau submissions.
- **Characteristics:** fixed formats and templates, hard deadlines, zero tolerance for error, full audit trail and lineage required.
- **Uses:** statutory returns, capital adequacy, provisioning, stress testing, audit.
- **Example:** monthly NPA classification and provisioning as per RBI IRAC norms must reconcile exactly with the general ledger; a mismatch invites regulatory penalty.

5.5 Other emerging sources

- **Alternative data:** GST returns, telecom/utility payments, e-commerce history for thin-file lending.
- **Unstructured data:** scanned documents, emails, call recordings, social media sentiment, news.
- **Machine/IoT data:** telematics for motor insurance (driving behaviour based premium).

6. Data Quality Dimensions

Data quality is “fitness for the intended use”. Five dimensions are prescribed in the syllabus; each one must have a measurable rule.

Dimension	Definition	How it is measured	BFSI example of failure
Accuracy	Value correctly reflects the real-world fact	% of records matching a trusted source (bureau, KYC document, GL)	Customer income recorded ₹50,000 but salary slip shows ₹25,000 → wrong loan eligibility
Completeness	All required fields and all expected records are present	% non-null per field; record count vs source count	PAN missing in 12% of accounts → CKYC/AML reporting fails
Consistency	Same fact agrees across systems and does not violate business rules	Count of cross-system mismatches and rule violations	Loan outstanding is ₹5.2 lakh in the loan system and ₹5.4 lakh in the GL
Timeliness (and currency)	Data available when needed and is up to date	Data latency; % of records refreshed within SLA	NAV file loads at 11 pm but the risk dashboard is used at 9 pm → yesterday's prices used
Validity (conformity)	Value follows the defined format, type, range and code list	% records passing format/range checks	Date of birth = 30-02-1990; IFSC of 10 characters; state code "XX" not in master

- **Additional dimensions used in industry:** **Uniqueness** (no duplicate entity), **Integrity** (valid relationships between tables), **Traceability/Lineage** (where the value came from). Uniqueness is often listed as the 6th core dimension.
- **Data quality scorecard:** assign a weight to each dimension per critical data element and publish a monthly score — for example “Customer PAN: completeness 99.1%, validity 98.4%, DQ score 98.7%”.
- **Trade-off:** higher timeliness (real-time feeds) often reduces accuracy because there is less time for validation — this trade-off must be decided consciously per use case.

Real-life example: Before RBI’s annual inspection, a bank runs a DQ scorecard on 40 critical data elements. Completeness of ‘borrower constitution’ is only 87%. Because Basel risk-weight depends on this field, incomplete data forces the bank to apply the harshest risk weight — consuming extra capital worth crores. Poor data quality here has a direct rupee cost.

7. Common Data Quality Issues in BFSI

7.1 Missing values

- **Types:** MCAR (missing completely at random), MAR (missing at random, explainable by other fields), MNAR (missing not at random — e.g. high earners refuse to declare income).
- **Typical BFSI causes:** optional fields in onboarding forms, failed system integration, legacy migration, customer refusal.
- **Detection:** null counts per column, missing-value heatmap, missingness by branch/date to find the faulty source.

- **Treatment options:** (i) delete the row only if very few and non-systematic; (ii) impute with mean/median/mode; (iii) impute by group (median income of that city and occupation); (iv) create a “Missing” category flag — often the flag itself is predictive; (v) go back to source and re-collect for regulatory fields where imputation is not allowed.
- **Rule for BFSI:** never impute a regulatory or identity field (PAN, Aadhaar reference, DOB). Imputation is acceptable for analytical features, not for compliance reporting.

Real-life example: In a home-loan dataset, ‘co-applicant income’ is blank for 40% of rows. Investigation shows it is blank when there is no co-applicant. Correct treatment is 0 plus a flag ‘has_co_applicant = No’, not the mean — otherwise eligibility is overstated for single-applicant loans.

7.2 Duplicates

- **Exact duplicates:** the identical row loaded twice (usually an ETL re-run).
- **Fuzzy / entity duplicates:** same person as “Ram Gopal Prajapati”, “R. G. Prajapati”, “Ramgopal Prajapati” with different address spellings.
- **Impact:** inflated customer count, wrong exposure per borrower (a serious credit-risk breach), double-counted premium income, duplicate marketing spend, AML blind spots.
- **Detection:** COUNT vs COUNT(DISTINCT) on the key; grouping by PAN/mobile/DOB; fuzzy matching (Levenshtein distance, Soundex), Power Query “Remove Duplicates”, Excel COUNTIF.
- **Treatment:** define a golden key (PAN or CKYC number), use survivorship rules (keep the most recently updated record and the most complete address), maintain a Master Data Management (MDM) hub.

Real-life example: A bank believed one borrower had ₹4 crore exposure. After de-duplication across three customer IDs, the true single-borrower exposure was ₹11 crore — above the internal limit. De-duplication is not cosmetic; it is risk management.

7.3 Outliers

- **Definition:** observation far away from the rest of the distribution.
- **Two kinds:** (a) **error outliers** — salary entered as ₹5,00,00,000 instead of ₹5,00,000 (extra zeros); (b) **genuine outliers** — an HNI customer who truly transacts ₹5 crore.
- **Detection methods:** box-plot / IQR rule (below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$), Z-score beyond ± 3 , percentile scan (P99), scatter plots, isolation forest for multivariate cases.
- **Treatment:** correct if it is a data error; **winsorise/cap** at P1-P99 for modelling; **transform** with log for skewed amounts; **segment separately** (HNI book analysed apart); and never delete a genuine outlier that is the business signal.
- **Golden rule in BFSI:** in fraud and AML analytics the outlier is the target — removing outliers would destroy the use case.

Real-life example: An insurer’s average claim size jumps 3× in one month. The cause is a single ₹2 crore fire claim. For the pricing model it is capped; for the reinsurance and reserving report it is kept in full and shown separately as a large-loss item.

7.4 Format Inconsistencies

- **Dates:** 01/02/2026 (DD/MM) vs 01/02/2026 (MM/DD), Excel serial numbers, text dates. Fix: one standard, ISO YYYY-MM-DD in the warehouse.

- **Numbers stored as text:** “₹1,20,000”, “1.2 Lakh”, “(5,000)” for negative, trailing spaces. Fix: strip symbols, convert type, define unit (always rupees, never mixed lakh/crore).
- **Codes and categories:** “M/F” vs “Male/Female”, “MH” vs “Maharashtra”, “SB” vs “Savings”. Fix: code master and mapping table.
- **Identifier formats:** mobile with +91 / 0 prefix, IFSC in lower case, PAN with spaces, account numbers with leading zeros lost by Excel.
- **Currency and unit mismatch:** one file in ₹ crore, another in ₹ thousand — silently wrong totals.
- **Case and whitespace:** “Indore ”, “indore”, “INDORE” counted as three cities.
- **Tools to fix:** Power Query (Change Type, Trim, Clean, Split/Merge Column, Replace Values), Excel TEXT/VALUE/DATEVALUE/TRIM, SQL CAST and CASE mapping.

Real-life example: A branch-wise deposit report showed Indore three times with different totals. The reason was trailing spaces and case differences in the branch name. After Trim + Upper in Power Query the three rows merged into one — the dashboard number changed materially.

7.5 Other frequent issues

- **Stale data:** customer address not updated for 7 years.
- **Referential integrity breaks:** a transaction pointing to a customer ID that does not exist in the customer master.
- **Truncation and encoding:** long names cut at 30 characters; special characters becoming “?”.
- **Reconciliation gaps:** dashboard total not tying to the general ledger — the fastest way to lose management trust in a dashboard.

8. Data Governance in Financial Services

- **Definition:** data governance is the framework of ownership, policies, standards, roles and controls that decides who can take what action on which data, when, and with what accountability.
- **Why it is non-negotiable in BFSI:** regulatory reporting accuracy (RBI/SEBI/IRDAI), Basel principles (BCBS 239 on risk data aggregation), privacy law (DPDP Act 2023), audit trail, and the fact that a data error becomes a financial loss.

8.1 Key roles

- **Chief Data Officer (CDO):** owns the enterprise data strategy.
- **Data Owner (business):** accountable for a data domain, e.g. Head of Retail Assets owns loan data.
- **Data Steward:** defines and maintains definitions, quality rules and issue resolution for the domain day-to-day.
- **Data Custodian (IT):** operates the systems, access control, backups and pipelines.
- **Data Consumer:** analysts and business users who must use certified data, not private copies.
- **Data Governance Council:** cross-functional forum that approves definitions, priorities and escalations.

8.2 Core components

- **Business glossary / data dictionary:** one agreed definition — e.g. “Active customer = at least one financial transaction in the last 90 days”. Stops the classic “your report says 12 lakh customers, mine says 9 lakh” fight.
- **Critical Data Elements (CDEs):** the limited list of fields that matter most (PAN, NPA flag, exposure, premium) — quality effort is concentrated here.
- **Data quality rules and scorecards:** automated checks per dimension with thresholds and owners.
- **Metadata and data lineage:** traceability from a dashboard number back to the source column and transformation. Regulators can and do ask for this.
- **Reference and master data management (MDM):** single golden record for customer, product, branch.
- **Access control and classification:** Public / Internal / Confidential / Restricted; role-based access; masking of PII; encryption at rest and in transit.
- **Privacy and consent:** purpose limitation, consent records (Account Aggregator framework), data retention and deletion schedule.
- **Issue management:** DQ issue log with severity, root cause, remediation owner and target date.
- **Audit and change control:** versioned definitions, approvals, and evidence for auditors.

8.3 Data management frameworks

Framework	What it provides	BFSI relevance
DAMA-DMBOK (DAMA wheel)	11 knowledge areas — governance, architecture, modelling, storage, security, integration, documents, reference & master data, DW/BI, metadata, data quality	The standard checklist used to design a bank’s data function
BCBS 239	14 principles for risk data aggregation and reporting — accuracy, completeness, timeliness, adaptability, governance	Mandatory expectation for large banks; drives lineage and CDE programmes
COBIT / ITIL	IT governance and service management controls	Access, change and incident control around financial systems
ISO 8000 / ISO 25012	Data quality standards and vocabulary	Formal definitions for DQ rules and measurement
DCAM / CMMI-DMM	Data management maturity assessment models	Board-level maturity scoring of the data programme
Regulatory: RBI Master Directions, DPDP Act 2023, Basel III, IRDAI & SEBI norms	Legal obligations on reporting, privacy, capital and disclosure	Define the hard constraints inside which analytics must operate

8.4 Modern data architecture terms (good to know)

- **Data warehouse:** structured, modelled, governed store for reporting (star schema — Unit 2).
- **Data lake / lakehouse:** raw and semi-structured data at low cost, used for exploration and ML.
- **ETL vs ELT:** transform before load vs load then transform (Power Query is a small-scale ETL tool).
- **Data mart:** subject-specific slice — e.g. a credit-risk mart.
- **Golden source / single source of truth:** the certified table that all dashboards must use.
- **Data mesh:** domain teams publish their data as products with their own quality contracts.

Real-life example: A private bank's Power BI 'NIM dashboard' is certified: it reads only from the governed finance mart, the definition of NIM is locked in the business glossary, the data steward signs off monthly, lineage is documented, and access is restricted by role. That is governance in practice — the same chart built from a personal Excel file downloaded to a laptop would fail audit.

9. Unit-1 Summary (Quick Revision)

- Business analytics = data + method + domain, ending in a **decision**.
- Four types: descriptive (what), diagnostic (why), predictive (what next), prescriptive (what to do).
- Lifecycle: problem definition → collection → processing → analysis → visualisation → decision → monitoring.
- Four BFSI data sources: transactional, market, customer, regulatory (plus alternative/unstructured).
- Five DQ dimensions: accuracy, completeness, consistency, timeliness, validity (+ uniqueness).
- Four common issues: missing values, duplicates, outliers, format inconsistencies — each with detection and treatment.
- Governance = roles (owner, steward, custodian) + glossary + CDEs + DQ rules + lineage + access control, guided by DAMA-DMBOK and BCBS 239.

10. Practice Questions (exam-oriented)

- Define business analytics and differentiate it from business intelligence with a BFSI example.
- Explain the four types of analytics with one banking and one insurance example each.
- Draw and explain the data analytics lifecycle for a credit-card fraud project.
- Compare transactional, market, customer and regulatory data on volume, frequency, ownership and use.
- List the five data quality dimensions and give one measurable rule and one BFSI failure for each.
- How would you detect and treat missing values, duplicates, outliers and format inconsistencies in a loan dataset? Justify each treatment.
- What is data governance? Describe the roles and components, and explain BCBS 239 in one paragraph.
- Practical (lab): take a raw loan CSV, profile it in Power Query, fix formats, remove duplicates, and publish a one-page data quality scorecard.